

6th June 2026

END to END using all the tools

Cloud → Infra → Hardware → Azure → Linux Base → application →

DevOps

CI and CD pipeline

CD → kubeadmn → single → Application run → Jenkins and ANsible →

Terrafrom

CI →

1. Windows or Linux → IDE Visual studio → Linux? OpenSource →
GPL → terraform → AWS terrafrom → pay → HCL → OpenTofu
2. Code → AI or Dev → SCM → Github or
3. Github Push code
4. Terrafrom
5. Jenkins poll
6. ANsible → Server Azure
7. Kubeadm →
8. Prometheus Monitoring

DevOps → Manual

Manual

Automation Automation and automation →

AI for DevOps

Jan → June → Fund

Project

Company → Task →

Senior Lead →

Amazon Q

Co

Server —> ready to on local

IDE —> everything —>

CI —>

Azure VM —> SSH code deployment

LLM or Deployment —> Dockerized —> Kubernetes

CI

Or

CD—> Kubernetes

AWS EC2 —> Azure Virtual Machine —> GCP Compute Engine —> Digital Ocean
—> Droplet

Insance type —> ts

Azure B

Dockerfile —>

FREE Container

FROM

RUN

RUN

RUN

RUN

EXPOSE

ENTRYPOINT

CMD

Fork → OpenSource → Make changes →
CLone →

AWS → Big → Search

Azure →
Resource Group → DevOps RG → Billing →

7th June 2026

CD

1. Jenkins
2. K8s
 1. Deployment.yml
 2. Services.yml
 1. NodePort

K8s

Add-hoc → ls

Services → Systemctl status Jenkins → 24/7 365

Terraform → EKs cluster

```
export ARM_CLIENT_ID="your-azure-service-principal-app-id"
export ARM_CLIENT_SECRET="your-azure-service-principal-secret-key"
export ARM_SUBSCRIPTION_ID="your-azure-subscription-id"
export ARM_TENANT_ID="your-azure-tenant-id"
```

terraform usage

Value "Mzg8Q~MnFqfDoF-3WgqsGMcvyn-ooO7YhxT0gct8"
Secret ID "00d97185-4d91-43d0-a061-ac410b43598b"

```
export ARM_CLIENT_ID="Mzg8Q~MnFqfDoF-3WgqsGMcvyn-ooO7YhxT0gct8"
export ARM_CLIENT_SECRET="00d97185-4d91-43d0-a061-ac410b43598b"
```

```
export ARM_SUBSCRIPTION_ID="cd459aae-8e9d-4069-8ab8-fd60d11b3e04"  
export ARM_TENANT_ID="3be84d67-b46a-4959-bfe8-00e8d3fc27e7"
```

az login

az account set --subscription "cd459aae-8e9d-4069-8ab8-fd60d11b3e04"

RCA →

Terraform → IAM

az login → login session in the chrome →

Login

Terraform code

Ready made → 100%

RG

Network

Subscription

POA

Eks Cluster

Kubeadm

Docker engine

One sample application

Deployment.yml → Pod docker image

Nodeport.yml

[GitHub.com](https://github.com)

Jenkins + Eks cluster

SSH → Configuration EKs cluster

Kubectl

Jenkins

CI —> to build the image push to docker hub

Github token —> Jenkins credential

Pipeline build docker image and push to docker hub

dckr_pat_-UUSW5sjnSdYNvMe65KcaA82bnk

K8s Server Execution data on

testadmin@prod-eks-cluster-vm:~\$ history

```
1 sudo -i
2 df -h
3 sudo hostnamectl hostname prod-eks-cluster-vm
4 exit
5 sudo -i
6 clear
7 ls
8 sudo apt update -y
9 sudo apt install docker.io
10 htop
11 sudo systemctl status docker
12 sudo systemctl stop docker
13 sudo systemctl stop docker.socket
14 sudo systemctl status docker
15 sudo systemctl stop docker
16 sudo apt update
17 sudo apt install -y containerd
18 sudo mkdir -p /etc/containerd
19 containerd config default | sudo tee /etc/containerd/config.toml
20 sudo systemctl restart containerd
21 sudo systemctl enable containerd
22 sudo systemctl status containerd
23 sudo swapoff -a
24 sudo sed -i ' / swap / s/^/#/' /etc/fstab
25 htop
26 cat <<EOF | sudo tee /etc/modules-load.d/k8s.conf
27 overlay
28 br_netfilter
29 EOF
30 sudo modprobe overlay
```

```
31 sudo modprobe br_netfilter
32 cat <<EOF | sudo tee /etc/sysctl.d/k8s.conf
33 net.bridge.bridge-nf-call-iptables=1
34 net.bridge.bridge-nf-call-ip6tables=1
35 net.ipv4.ip_forward=1
36 EOF
37 sudo sysctl --system
38 sudo apt-get update
39 sudo apt-get install -y kubelet kubeadm kubectl
40 sudo apt-mark hold kubelet kubeadm kubectl
41 sudo mkdir -p /etc/containerd
42 containerd config default | sudo tee /etc/containerd/config.toml
43 sudo sed -i 's/SystemdCgroup = false/SystemdCgroup = true/' /etc/
containerd/config.toml
44 sudo apt update
45 sudo apt install -y apt-transport-https ca-certificates curl gpg
46 curl -fsSL https://pkgs.k8s.io/core:/stable:/v1.34/deb/Release.key | sudo
gpg --dearmor -o /etc/apt/keyrings/kubernetes-apt-keyring.gpg
47 echo 'deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg]
https://pkgs.k8s.io/core:/stable:/v1.34/deb/ /' | sudo tee /etc/apt/sources.list.d/
kubernetes.list
48 sudo apt update
49 sudo apt install -y kubelet kubeadm kubectl
50 sudo apt-mark hold kubelet kubeadm kubectl
51 sudo kubeadm init --pod-network-cidr=192.168.0.0/16
52 kubectl get pod
53 sudo kubectl get pod
54 mkdir -p $HOME/.kube
55 sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
56 sudo chown $(id -u):$(id -g) $HOME/.kube/config
57 kubectl get nodes
58 kubectl apply -f https://raw.githubusercontent.com/projectcalico/calico/
v3.30.3/manifests/calico.yaml
59 kubectl get nodes
60 kubectl get pods -A
61 kubectl get nodes
62 kubectl get pods -A
63 htop
64 kubectl get pods -A
65 htop
66 kubectl get nodes
67 kubectl create deployment nginx --image=nginx
68 kubectl expose deployment nginx --port=80 --type=NodePort
69 kubectl get svc
70 exit
71 kubectl get svc
72 exit
```

```
73 kubectl get pod
74 kubectl describe pod nginx-66686b6766-bf9tv
75 kubectl describe node $(kubectl get nodes -o name | cut -d/ -f2) | grep
Taints
76 kubectl taint nodes --all node-role.kubernetes.io/control-
plane:NoSchedule
77 kubectl taint nodes --all node-role.kubernetes.io/control-plane
78 kubectl taint nodes --all node-role.kubernetes.io/control-
plane:NoSchedule
79 kubectl taint node prod-eks-cluster-vm node-role.kubernetes.io/control-
plane:NoSchedule-
80 kubectl get pod
81 exit
82 kubectl get node
83 kubectl get svc
84 exit
85 kubectl get node
86 kubectl get pod
87 kubectl describe pod nfts-deployment-7c9496f64b-ghqz7
88 kubectl logs nfts-deployment-7c9496f64b-ghqz7
89 exit
90 kubectl rollout restart deployment/nfts-deployment
91 kubectl get pod
92 kubectl logs nfts-deployment-6c46c9d5f4-vtfm4
93 kubectl get deployment
94 kubectl delete deployment nfts-deployment nginx
95 kubectl get deployment
96 kubectl get svc
97 kubectl delete svc nfts-service nginx
98 kubectl get svc
99 kubectl get deployment
100 kubectl get pod
101 kubectl pod nfts-deployment-7c9496f64b-mj2kc
102 kubectl log nfts-deployment-7c9496f64b-mj2kc
103 kubectl logs nfts-deployment-7c9496f64b-mj2kc
104 kubectl delete svc nfts-service nginx
105 kubectl get svc
106 kubectl get pod
107 kubectl logs nfts-deployment-7c9496f64b-mj2kc
108 exit
109 docker images
110 sudo -i
111 kubectl set image deployment/nfts-deployment nfts-
appcontainer=devopsdecode/jan-end-to-end-project:v1 --record
112 kubectl get pod
113 kubectl get deployment
114 kubectl delete deployment nfts-deployment
```

```
115 kubectl get deployment
116 kubectl get pods -l app=nfts-app
117 exit
118 kubectl set image deployment/nfts-deployment nfts-
appcontainer=devopsdecode/jan-end-to-end-project:v2 --record
119 kubectl get pod
120 kubectl get deployment
121 exit
122 kubectl po
123 kubectl get deployment
124 kubectl delete deployment nfts-deployment
125 kubectl get deployment
126 kubectl get svc
127 kubectl delete svc nfts-service
128 kubectl get svc
129 kubectl get deployment
130 exit
131 kubectl get pod
132 kubectl get svc
133 exit
134 kubectl get pod
135 exit
136 kubectl get deployment
137 kubectl delete deployment nfts-deployment
138 kubectl get svc
139 kubectl delete svc nfts-service
140 exit
141 kubectl get svc
142 kubectl get deployment
143 kubectl get svc
144 kubectl get pod
145 kubectl get svc
146 kubectl describe deployment nfts-deployment-65dfcb6f66-n5l89
147 kubectl describe deployment nfts-deployment
148 kubectl get svc
149 exit
150 kubectl get pod
151 kubectl get deployment
152 exit
153 curl https://raw.githubusercontent.com/helm/helm/main/scripts/get-
helm-3 | bash
154 helm version
155 helm repo add prometheus-community https://prometheus-
community.github.io/helm-charts
156 helm repo update
157 kubectl create namespace monitoring
158 helm install prometheus prometheus-community/kube-prometheus-stack
```



```
--namespace monitoring
159 helm list -n monitoring
160 kubectl get pods -n monitoring
161 helm list -n monitoring
162 kubectl get pods -n monitoring
163 kubectl get svc -n monitoring
164 kubectl patch svc prometheus-grafana -n monitoring -p '{"spec":
{"type":"NodePort"}}'
165 kubectl get svc -n monitoring
166 exit
167 history
testadmin@prod-eks-cluster-vm:~$
```

Jenkins executed data on server

azureuser@jan-end-to-end-project:~\$ history

```
1 sudo -i
2 exit
3 htop
4 sudo systemctl status jenkins
5 terraform --version
6 ansible --version
7 sudo systemctl status jenkins
8 sudo systemctl start jenkins
9 journalctl -xeu
10 journalctl
11 sudo systemctl stop jenkins
12 sudo systemctl status jenkins
13 sudo systemctl start jenkins
14 cd /var/log/jenkins/n
15 cd /var/log/jenkins/
16 ls -rlth
17 journalctl -xe
18 sudo apt remove jenkins
19 sudo apt install jenkins
20 sudo systemctl status jenkins
21 journalctl -xe
22 sudo apt remove openjdk-17-jre
23 sudo apt install openjdk-
24 sudo apt install openjdk-*
25 exit
```

```
26 sudo systemctl status jenkins
27 sudo -i
28 sudo systemctl status jenkins
29 sudo systemctl start jenkins
30 "journalctl -xeu jenkins.service
31 journalctl -xeu jenkins.service
32 sudo -i
33 sudo systemctl status jenkins
34 htop
35 sudo systemctl status jenkins
36 terraform -version
37 terraform --version
38 ansible --version
39 telnet localhost 8080
40 exit
41 sudo systemctl status jenkins
42 cat /var/lib/jenkins/secrets/initialAdminPassword
43 sudo cat /var/lib/jenkins/secrets/initialAdminPassword
44 exit
45 sudo apt install docker.io
46 docker version
47 docker ps
48 sudo usermod -aG docker jenkins
49 sudo systemctl restart jenkins
50 sudo su - jenkins
51 docker ps
52 history
azureuser@jan-end-to-end-project:~$
```

POA to complete this project

Day1

1. Planning new project
 2. CI jobs
 3. Laptop
 4. Build application
 5. Build docker image
 6. Scanned traviy
 7. Docker image
 8. CI completed
 9. CD AZure
 10. Installation of tools using ansible (local)
- At

Day2 :- CD

1. Using Terraform create the k8s cluster single
2. Tried to connect the Azure CLI using export and az login
3. **K8s cluster**
4. Deployment.yml and services.yml
5. Github pushed all the code
6. Jenkins job CI pipeline
7. Jenkins job CD pipeline
8. Problem image pull
9. BuildX
10. Research and development
11. Application end to end deployment using ci and cd
12. Monitoring using helm chart

diff b/w nordport and loadbalancer → 1+ EKS cluster → ALB → AWS or Azure → HA of ALB

Terraform vs ansible if both can do infra provisioning (edited)

Terraform → before login into the server

Ansible → After login into the server